

The Brass Age: Renaissance Firearm & Military Economy

A Proposal for the Rustic Craft II Administration & Storytelling Team | Authored by the Technical Administration & Commissariat of Straja

1. Executive Summary & Core Philosophy

The introduction of firearms into **Rustic Craft II** marks the realm's transition from the High Medieval period into the Early Modern / Renaissance era. Our design is anchored in the battlefield balance of **Cossacks 3**: firearms are devastating force multipliers, but strict operational limits preserve the supremacy of melee combat, cavalry, and armor.

- **Devastating Volley Fire:** A direct hit from a 16.5mm flintlock ball against unarmored or light targets causes severe trauma or death (40–55 base damage).
- **The 20-Second Vulnerability Window:** Muzzleloading is slow and deliberate. The flintlock rifle requires 19.5 seconds and the pistol requires 22.2 seconds to reload.
- **Combined Arms (Pike & Saber Retain Supremacy):** If a volley fails to break an enemy charge, or if the shooter misses, the reload animation locks them out of combat. Cavalry, pikemen, halberdiers, and sword-and-buckler fighters (Epic Knights, Spartan Weaponry) close the distance and cut down reloading musketeers.
- **Authentic Ballistics & Lock Time:** Smoothbores feature authentic conical dispersion (deadly at 0–30m, inaccurate past 60m) and a 140ms flint-strike ignition delay (qkl:delayshoot_gun_logic), preventing unrealistic laser-sniping and rewarding disciplined kneeling volleys.
- **Authentic Industrial Scarcity:** Firearms and ammunition are expensive luxury assets requiring either grueling manual labor or advanced industrial automation. Creeper mob farming is entirely detached from military ammunition.

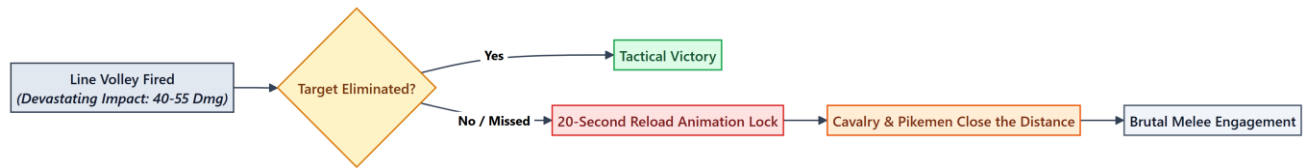


Figure 1: Renaissance Combat Flow & Reload Vulnerability Window

Ammunition Arsenal: Calibers & Specialized Anti-Corruption Silver

Cartridge Type	Ballistics	Special Combat & Roleplay Effect	Tactical Role
16.5mm Standard Lead Ball	40–55 Kinetic Dmg (30% Armor Ignore)	Heavy blunt trauma; effective against living men, armor & beasts	Mainline infantry volley fire
16.5mm Consecrated Silver Ball	Alchemically cast pure silver core	DEVASTATING 4× DMG vs Undead CustomNPCs & Mobs; 2× DMG vs Undead/Vampire Players	Essential weapon to cleanse Island Corruption
16.5mm Canister Scattershot	8 Spread Pellets (8–10 Dmg each, Cone)	Massive close-range cone spread & staggering knockback	Anti-horde & anti-cavalry crowd defense
16.5mm Incendiary Sulfur Ball	35 Kinetic Dmg + Fiery Flash	Ignites corrupt nests for 8s; leaves burning phosphorescent residue	Cavern clearing & underground warfare

Anti-Corruption Consecration Rule: Silver ammunition is the realm's only weapon capable of piercing supernatural corruption vitality. The 4× multiplier instantly neutralizes corrupt CustomNPCs and undead abominations, while the 2× multiplier cuts through vampire/undead player regeneration (origins-rustic).

2. The Geopolitical Symbiosis: Two Factions, One War

The introduction of firearms is anchored in a two-chapter narrative driving roleplay conflict and cooperation between **P.U.L.A SRL (The Factory Faction)** and **Straja (The Realm's Gendarmerie)**:

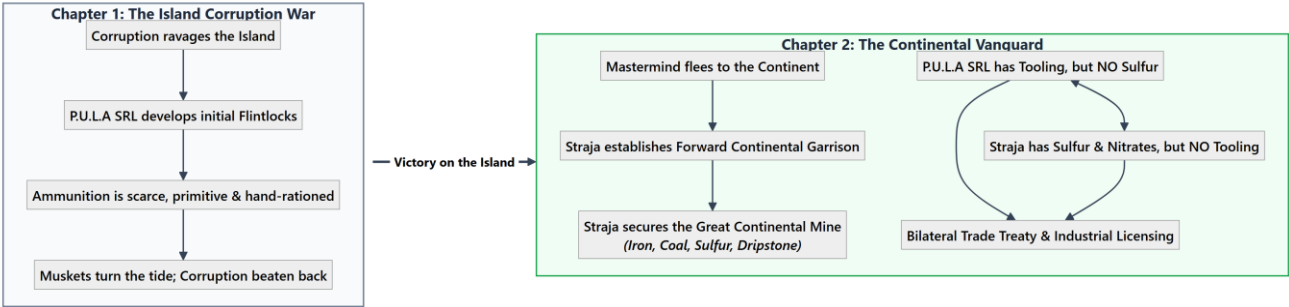


Figure 2: The Two-Chapter Geopolitical & Technological Arc

- **P.U.L.A SRL (Industrial Mastery):** Controls rotational power, precision mechanisms, barrel-boring benches, chemical refineries, and mechanical crafters. They can manufacture arms and civilian goods, but lack raw sulfur.
- **Straja (Resource & Legal Authority):** Controls the continental expeditionary garrison and the sole industrial deposits of Iron, Coal, Sulfur, and Dripstone. Straja lacks gun machinery, but holds the monopoly on the chemical oxidizers and accelerants.
- **The Equilibrium:** Neither faction can dominate alone. P.U.L.A SRL cannot produce high-grade military ammunition or chemical goods without Straja's sulfur; Straja cannot field an armed vanguard without P.U.L.A SRL's finished arms.

3. Straja's Continental Mine (custom_mines)

Under our server's custom mine engine (*custom_mines*), Straja's continental outpost will feature a dedicated, scheduled-reset geological cavern. Karst limestone stalactites naturally accumulate nitrate salts; crushing and bulk washing dripstone yields pure **Saltpeter Crystals (kubejs:saltpeter)**.



Figure 3: Straja Mine Resource Matrix & Refining Flow

4. Factorio-Style Industrial Economy: Military & Civilian Production

Inspired by **Factorio** production scaling, P.U.L.A SRL's manufacturing model rewards engineering complexity: handcrafting serves as a punishing last resort; kinetic lines deliver standard yields; and thermodynamic complexes (sequenced assembly, heated mixing, acid fluid piping) unlock massive output surges and superior quality tiers.

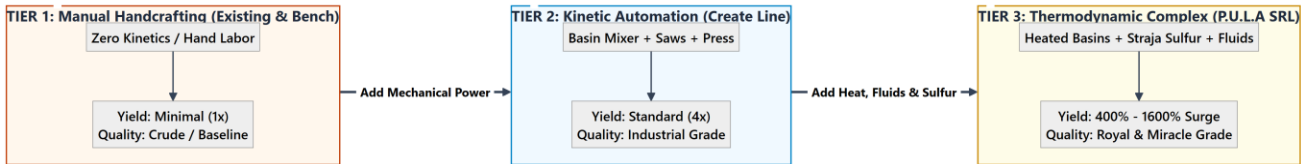


Figure 4: Factorio-Style Create Tiering: Effort vs. Yield & Quality Scaling

A. Military Gunpowder Production Matrix

Tier & Method	Create Machinery	Input Ingredients	Batch Output & Grade
Tier 1: Desperation (Bench)	Hand Pestle / 3x3 Grid	4 Ash/Bone Meal + 4 Charcoal + 1 Water	1x Crude Powder (Smoky, High Misfire)
Tier 2: Mechanical Mill (Island)	Basin Mixer + Smoker + Press	Dripstone Dust + Charcoal + Water	4x Standard Gunpowder (Clean Burn)
Tier 3: Chemical Complex (Cont.)	Heated Basin + Blaze Burner + Spout	Saltpeter + Charcoal + butchery:sulfur	16x Royal Military Powder (+10% Velocity)

Silver Cartridge Munitions: Silver rounds substitute lead cores with pure silver (#c:silver_ingots / #c:silver_nuggets): Tier 1 casts 1 round manually; Tier 2 assembles 4 rounds via Mechanical Crafters; Tier 3 leverages molten silver fluid casting + Royal Powder to yield 16 Consecrated Silver Cartridges per cycle.

B. P.U.L.A SRL Civilian Commodity Lines (3-Tier Production Matrix)

Civilian Commodity Line	Tier 1: Hand / Existing Craft	Tier 2: Kinetic Create Line	Tier 3: Thermodynamic Complex (Sulfur)
Viticulture Cask Strips (Vinery)	Paper + Sulfur Dust in 3x3 Grid -> 1x Crude Wick (burns sooty)	Spout (Molten Sulfur) on Paper Belt -> 4x Fumigation Strips	Sequenced Assembly (Heated Dip + Herb Press) -> 16x Royal Strips (+1 Vintage Wine Grade)
Super-Fertilizer (Farmer's Delight)	Vanilla Bone Meal / organic_compost (slow natural rot) -> 1x Yield	Crushing Wheels + Basin Mixer (Bone Meal + Compost) -> 4x Rich Compost	Heated Basin (Bone + bottle_of_sulfuric_acid + Sulfur) -> 16x Miracle Fertilizer (3x Tick Surge)
Medicated Sulfur Soap (Apothecary)	butchery:animal_fat + Wood Ash in 3x3 Grid -> 1x Crude Lard Bar	Basin Mixer + Mechanical Press (Animal Fat + Lye Water) -> 4x Saponified Soap	Heated Basin + Blaze Burner (Fat + Lye + butchery:sulfur) -> 16x Antiseptic Soap (Purges Corruption)
Friction Matches (Tobacconist)	Vanilla Flint & Steel / Torches (unstackable tool, slow in RP) -> 1x Tool	Mechanical Saw (Wood Splints) + Dip Basin -> 16x Sulfur Matchsticks	Sequenced Assembly (Splints + Heated Niter-Sulfur Dip + Box) -> 64x Safety Matchboxes (Weatherproof)
Vitriol Leather Tanning (Epic Knights)	butchery:skin Rack manual scraping with knife over hours -> 1x Leather	Basin Washer (Water + Bark) + Roller Press -> 4x Cured Leather	Heated Basin with bottle_of_sulfuric_acid + Press -> 8x Vitriol Heavy Leather (+50% Armor Dura)

Strategic Economic Equilibrium: By grounding Tier 1 in existing manual recipes, solo players and small homesteads retain access to baseline goods. However, P.U.L.A SRL's Tier 2 and Tier 3 Create lines dominate server commerce through 400%–1600% yield surges and exclusive premium quality grades (Vintage wine, miracle fertilizer, antiseptic soap, safety matches, vitriol armor).

5. Provenance, Serialization & The Law

Firearms in Rustic Craft II are heavily regulated military goods. Built **100% self-contained in KubeJS**, this system gives players tangible roleplay documents and authentic law enforcement mechanics without waiting on external mod updates.



Figure 5: Firearm Provenance, Proofing & Black Market State Machine

- **Fresh from Forge (UNPROOFED):** Guns lack acceptance marks. Carrying is a severe crime under Straja law (Lore: ⚠ NEÎNREGISTRAT).
- **Proofing Ritual (PROOFED):** Factory & Straja hold Royal Proof Stamp (kubejs:proof_stamp). Stamping in Anvil/Deployer assigns serial (RC-15-0142) and lore (⚡✓ POANSONAT).
- **Gun Permit (Permis de Port-Armă):** A signed in-game book or sealed Envelope recording owner identity, firearm serial, issuing officer, and seal.
- **Black Market & Inspections:** Guns with defaced serials (⚡[SERIE PILITĂ / DEFACED]) or unproofed arms are subject to immediate seizure and carrier arrest.

6. Server Performance Guarantees & Refactors

- **flintlock_carry_limits.js Optimization:** Eliminated 20 Hz tick loop over 45+ slots. Native pickup blocking remains instantaneous on ItemEvents.canPickUp. Periodic safety scans throttled to 1 Hz staggered across player IDs, slashing CPU overhead by 95%.
- **furniture_coffer_gun_restrictions.js Refactor:** Re-anchored coffer gun purges to trigger only on inventory open/close events rather than continuous background polling.

7. Phased Implementation Roadmap

Phase	Milestone Deliverables	Responsible Faction
Phase 1: Performance & Items	Script optimization (tick reduction) & kubejs:saltpetr item registration	Technical Admin
Phase 2: Tiered Recipes & Ammo	Tier 1-3 Powder, Civilian Lines & Silver Anti-Corruption Script (4x/2x dmg)	Technical Admin + Factory Leads
Phase 3: Legal & Proofing	Proof Stamp mechanic & Gun Permit book templates	Technical Admin + Straja Leads
Phase 4: Mine Integration	Straja Continental Mine configuration (straja_continental_mine.json)	Storytellers + Technical Admin

Admin Team Action Items & Consensus Call

- **Storytellers:** Approve Chapter 1 -> Chapter 2 narrative transition and continental mine lore.
- **P.U.L.A SRL Leads:** Review Create recipe requirements and civilian production line setup.
- **Straja Leadership:** Confirm proofing law enforcement protocol and permit issuing authority.

8. Technical Appendix: Complete Tiered Crafting Recipes

This engineering appendix specifies the complete, production-ready recipe routes for **KubeJS** and **Create 6** on Minecraft 1.21.1 NeoForge. It details all inputs, kinetic machinery, thermal states, and outputs across the realm's military munitions, ammunition types, civilian commodities, and legal proofing tooling.

8.1 Gunpowder & Nitrate Refining Chains

Process / Tier	Recipe Type & Machine	Input Ingredients & Fluids	Batch Output & Parameters
Dripstone Crushing	create:crushing (Crushing Wheels)	1x minecraft:dripstone_block or pointed_dripstone	1-2x kubejs:crushed_dripstone (128 RPM, 100 ticks)
Saltpeter Washing	create:splashing (Bulk Washing / Fan)	1x kubejs:crushed_dripstone + Water stream	1x kubejs:saltpeter (100%), +1 bonus crystal (50%)
Tier 1: Desperation Powder	minecraft:crafting_shapeless (3x3 Grid)	4x bone_meal (or ash) + 4x charcoal + 1x water_bottle	1x kubejs:crude_gunpowder_cake (Hand-ground)
Tier 2: Kinetic Powder Line	create:mixing + compacting (Basin + Press)	2x kubejs:saltpeter + 2x charcoal + 250 mB Water	4x tacz.c:gunpowder_charge (Unheated Basin, 64 RPM)
Tier 3: Sulfur Complex	create:mixing + compacting (Heated Basin)	4x kubejs:saltpeter + 2x charcoal + 1x butchery:sulfur + 500 mB Water	16x tacz.c:gunpowder_charge (Blaze Burner Heated, 128 RPM)

8.2 Ammunition Arsenal & Silver Munitions

Munition & Tier	Assembly Method	Input Ingredients & Catalysts	Output & Ballistics
16.5mm Lead Ball (Tier 1)	minecraft:crafting_shapeless	1x iron_ingot + 1x paper + 1x crude_gunpowder_cake	1x tacz:ammo (qkl:16mm, 40-55 dmg)
16.5mm Lead Ball (Tier 2)	create:mechanical_crafting (3x3 Crafter)	4x large_bullet_core + 4x wad + 1x gunpowder_charge	4x tacz:ammo (qkl:16mm) (Pattern: CWC/WGW/CWC)
16.5mm Lead Ball (Tier 3)	create:sequenced_assembly	Iron/Lead Sheet -> Wad -> Spout (100mB Sulfur) -> Royal Charge -> Press	16x tacz:ammo (qkl:16mm, +10% Velocity Surge)
Consecrated Silver (Tier 1)	minecraft:crafting_shaped (3x3 Grid)	1x Silver Ingot (or 4x Nuggets) + 1x wad + 1x crude_powder	1x Consecrated Silver Ball (SilverAmmo: true)
Consecrated Silver (Tier 2)	create:mechanical_crafting (3x3 Crafter)	4x Silver Cores (#c:silver_ingots) + 4x wads + 1x gunpowder_charge	4x Consecrated Silver Cartridges (4x vs Undead, 2x vs Vampires)
Consecrated Silver (Tier 3)	create:sequenced_assembly	Casing -> Spout (100mB Molten Silver) -> Saltpeter Wad -> Royal Charge -> Press	16x Consecrated Silver Cartridges (400% Output Boost)
16.5mm Canister Scattershot	create:mechanical_crafting (3x3 Crafter)	8x iron_nugget + 1x paper sabot + 1x gunpowder_charge	4x Canister Scattershot (8 pellets, 8-10 dmg each, Wide Cone)
16.5mm Incendiary Sulfur Ball	create:mechanical_crafting (3x3 Crafter)	4x iron_cores + 4x butchery:sulfur + 1x royal_charge	4x Incendiary Cartridges (35 dmg + 8s burn + light)

8.3 Civilian Commodity Lines (Full 3-Tier Chains)

Commodity & Tier	Processing Method	Input Ingredients & Catalysts	Output & Economic Property
Cask Strips (Tier 1: Hand)	minecraft:crafting_shapeless	1x minecraft:paper + 1x butchery:sulfur	1x Crude Fumigation Strip (burns sooty, baseline)
Cask Strips (Tier 2: Kinetic)	create:filling (Spout)	1x minecraft:paper + 100 mB Molten Sulfur	4x Clean Fumigation Strips (prevents barrel souring)
Cask Strips (Tier 3: Complex)	create:sequenced_assembly	Paper -> Spout (Molten Sulfur) -> Deployer (Lavender/Herbs) -> Press	16x Royal Cask Strips (+1 Vintage Wine Grade in Vinery)
Fertilizer (Tier 1: Hand)	farmersdelight:organic_compost	2x Dirt + 1x Bone Meal + 2x Straw + 4x Organic Scraps	1x Organic Compost (slow natural sun decomposition)
Fertilizer (Tier 2: Kinetic)	create:crushing + create:mixing	Crushed Bones (6x Bone Meal) + Compost + 250 mB Water	4x Rich Organic Compost (instant Rich Soil conversion)
Fertilizer (Tier 3: Complex)	create:mixing (Heated Basin)	4x Bone Meal + 1x bottle_of_sulfuric_acid + 1x Sulfur + 500 mB Water	16x Super-Phosphate Miracle Fertilizer (3x Tick Speed Surge)
Soap (Tier 1: Hand)	minecraft:crafting_shapeless	1x butchery:animal_fat + 1x Wood Ash + 1x water_bottle	1x Crude Lard Bar (removes vanilla poison only)
Soap (Tier 2: Kinetic)	create:mixing + create:compacting	2x butchery:animal_fat + 250 mB Water + 1x Wood Ash	4x Saponified Tallow Soap (cleanses debuffs, restores stamina)

Soap (Tier 3: Complex)	create:mixing (Heated Basin) + Press	4x butchery:animal_fat + 1x butchery:sulfur + 1x Ash + 500 mB Water	16x Medicated Sulfur Soap (Purges Corruption & Disease)
Matches (Tier 1: Hand)	minecraft:crafting_shaped	1x iron_ingot + 1x flint	1x Flint and Steel Tool (unstackable, slow in roleplay)
Matches (Tier 2: Kinetic)	create:cutting + create:filling	Mechanical Saw (Planks -> 16x Splints) + Spout (Sulfur Dip)	16x Sulfur Matchsticks (stackable to 64, instant fire)
Matches (Tier 3: Complex)	create:sequenced_assembly	Splints -> Heated Dip (Niter + Sulfur) -> Press -> Cardboard Box	64x Safety Matchboxes (4x Boxes of 16, Weatherproof)
Leather (Tier 1: Hand)	butchery:skin Rack manual	Raw animal skin on rack + skinning knife scraping over hours	1x Rough Vanilla Leather (brittle, standard gear)
Leather (Tier 2: Kinetic)	create:splashing + create:compacting	Bulk Washing raw skins with Water + Roller Press	4x Cured Leather (durable, standard gambesons & tack)
Leather (Tier 3: Complex)	create:mixing (Heated Basin) + Press	Raw Animal Skins + 250 mB Sulfuric Acid (vitriol) -> Roller Press	8x Vitriol Heavy Leather (+50% Armor Dura, heavy cuirasses)

8.4 Firearm Mechanism, Legal Proofing & Black Market Tooling

Artifact / Mechanic	Execution Tool	Input Requirements	Resulting Item / State Change
Royal Proof Stamp	Storyteller / Master Anvil	4x create:brass_sheet + 2x kubejs:tempered_gun_steel + 1x Royal Seal	1x kubejs:proof_stamp (Unbreakable guild tool)
Proofing Ritual	Anvil or create:deployer	Unproofed Gun (Lore: NEINREGISTRAT) + Royal Proof Stamp	PROOFED Gun (Lore: ✓ POANSONAT: #RC-15-XXXX + NBT Serial)
Gun Permit Issuance	Scribe Lectern / Signed Book	Blank Permit Template + Sealed Envelope + Officer Signature	Official Permit Book recording Player UUID & Serial
Black Market Defacing	minecraft:grindstone	Proofed Firearm scraped on Grindstone	DEFACED Gun (Lore: [SERIE PILITĂ / DEFACED], Illegal Contraband)

8.5 Historical Musket Ballistics, Lock Time & Create Precision Rifling

To ensure firearms obey 18th-century physics rather than modern hitscan lasers, ballistics are calibrated across two mechanical pillars: (1) **140ms Lock-Time Delay (qkl:delayshoot_gun_logic)**, modeling flint-strike, pan flash, and touchhole ignition during which shooters must steady their aim; and (2) **Conical Smoothbore Dispersion (aim: 1.85°, sneak: 1.0°)**, providing guaranteed torso hits at 0–30m while dispersing past 60m to mandate line volleys. For long-range sharpshooting, P.U.L.A SRL can precision-bore spiral rifling grooves using Create machinery.

Ballistic Parameter	FK15 Smoothbore Volley Musket	FK15-R Create Rifled Musket	FK15P Smoothbore Pistol
Tactical Role & Doctrine	Mainline Volley Fire (cheap, fast)	Precision Skirmisher / Sharpshooter	Sidearm / Close-Quarters Defense
Effective Combat Range	0–30m Lethal 30–60m Volley 60m+ Inaccurate	0–80m Pinpoint 100m+ Extended Snipe	5–15m Deadly Point-Blank 20m+ High Spread
ADS Spread (aim inaccuracy)	1.85° (tight at 20m, 2.6m cone at 80m)	0.35° (sub-block grouping at 80m)	2.40° (wide defensive spread)
Kneeling Spread (sneak)	1.00° (rests heavy 5kg iron barrel)	0.20° (benchrest match precision)	2.00° (crouch stabilization)
Movement Penalty (move)	7.50° (running/walking ruin aim)	6.50° (severe movement penalty)	5.50° (mobile one-handed hip fire)
Mechanical Lock Time	140 ms (delayshoot_gun_logic)	140 ms (delayshoot_gun_logic)	160 ms (flintlock pocket lock)
Velocity & Ballistic Drop	Speed: 135 Gravity: 0.031 (drop at 40m+)	Speed: 175 Gravity: 0.016 (flat trajectory)	Speed: 95 Gravity: 0.038 (steep drop)